

The neighbourhood of a Christoffel word contains no Collatz cycle words

Working notes

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Abstract

A nontrivial positive cycle of the accelerated Collatz map with r odd steps and A halvings has a valuation word v whose Böhm–Sontacchi numerator $B(v)$ is divisible by $q = 2^A - 3^r$. Knight proved that the balanced word, the Christoffel word $\text{chr}(r, A)$, is not the word of a nontrivial cycle. We prove that, for every pair (r, A) with $r \geq 2$ and $2^A - 3^r > 1$, no word obtained from $\text{chr}(r, A)$ by one local move (an adjacent swap or a unit slide to a neighbour) satisfies $q \mid B(v)$, apart from the trivial-cycle word at $A = 2r$. The proof uses no Baker-type bounds; its only Diophantine input is a formally verified effective irrationality measure for $\log_2 3$. We also exclude a unit slid any distance (for $r \geq 40901$, which every nontrivial cycle satisfies), several two- and few-move families, and large classes of words when $\gcd(A, r) \geq 2$. Each result also holds for actual cycles. All statements have been formally verified in Lean 4. These are results about restricted cycle words. They do not exclude nontrivial cycles in general.

1 Setting

We use the accelerated map on odd integers, $x \mapsto (3x + 1)/2^{v_2(3x+1)}$. A cycle through odd $x_0, \dots, x_{r-1}$ has valuations $a_j = v_2(3x_j + 1) \geq 1$, partial sums $A_j = a_0 + \dots + a_{j-1}$ (with $A_0 = 0$) and $A = A_r$. The same cycle, run under the Terras map $T(n) = n/2$ (n even), $(3n + 1)/2$ (n odd), has length $L = A$ with r odd steps. Under the classical map $C(n) = n/2$ or $3n + 1$, it has length $L + r$; we call this a C -cycle.

Definition 1. For a word $v = (v_0, \dots, v_{r-1})$ of positive integers with partial sums A_j , put

$$B(v) = \sum_{j=0}^{r-1} 3^{r-1-j} 2^{A_j}, \quad q = 2^A - 3^r.$$

The (*lower*) *Christoffel word* of slope A/r is $\text{chr}(r, A)_j = \lfloor (j+1)A/r \rfloor - \lfloor jA/r \rfloor$, so that its partial sums are $\lfloor jA/r \rfloor$. Other papers use upper Christoffel words of binary parity words; these agree with ours up to rotation and convention. A word is *one move* from w if it arises from w by swapping two cyclically adjacent entries, or by moving one unit from an entry ≥ 2 to a cyclic neighbour (`NormGoal.OneMove`). The words one move from chr form its *radius-1 neighbourhood*.

Since the entries of chr differ by at most one, every nontrivial swap of chr is also a unit slide. A move of chr away from the wrap-around position changes exactly one partial sum

by ± 1 . We call it a *left flip* if that partial sum is lowered and a *right flip* if it is raised. In Mghirbi's language, the *defect area* E of a word is the total amount by which its partial sums lie below the Christoffel line, at a suitable rotation. Left flips have $E = 1$, whereas right flips can have E of order r .

The cycle equation (Böhm–Sontacchi) reads $qx_0 = B(v)$. A cycle word therefore satisfies $q \mid B(v)$, and this divisibility is invariant under rotation of v .

Prior work.

- *The Christoffel word is extremal.* Halbeisen–Hungerbühler proved this [1], and Fernández–Ibáñez revisited it [6].
- *The Christoffel word is not a nontrivial cycle word.* This is Knight, “Collatz high cycles do not exist” [2], in the coprime setting.
- *Lebel* [3] proved $\gcd(B(\text{chr}), D) = 1$ at radius 0 using a resultant (Theorem 4.2). He gave the local flip identity (Lemma 5.1), reduced radius 1 to the trinomials $X^s - X + 1$ prime by prime (Theorem 6.3), closed the canal $s = 2$, and conjectured radius-1 exclusion in a single-prime form (Conjecture 13.1). Here and below, theorem numbers refer to the v3 preprint.
- *Mghirbi* [4] used a distinguished unit ξ modulo $D = q$ with $\xi^p = \frac{1}{2}$, $\xi^K = \frac{1}{3}$, which is our g^{-1} , together with a rotation-numerator lift. He noted that the case $E = 0$ gives a three-line proof of Knight's theorem. For actual cycles he excluded the coprime range $E \leq (1.5359 \dots - o(1))p^{2/3}$ (with $p = r$), and the non-coprime cases $E = 1, 2$ (Props. 9.1, 9.2). This covers the left flips.
- *Solomon* [5], for actual cycles, excluded the one-swap words of left-flip type $(01 \mapsto 10)$. He used Laurent and Eliahou in the coprime case and a geometric-cofactor argument when $\gcd > 1$ (Prop. 6.3). His Prop. 6.2 also excludes the mechanical (Christoffel) word when $\gcd > 1$, by reducing to its primitive block.
- *A Collatz World forum post* [7] proves the prefix identity $B(01A) - B(10A) = 3^{|A|}$ for binary parity words and relates it to Christoffel words.

The other half of the radius-1 neighbourhood, the right flips, has large defect area. As far as we know, it is excluded neither by Mghirbi nor by Solomon. Lebel's closure of the canal $s = 2$ covers some right flips.

2 The main theorem

Theorem 2 (NormAll.no_cycle_one_move_christoffel_all). *Let $r \geq 2$ and $3^r + 1 < 2^A$. If v is one move from $\text{chr}(r, A)$ and $v \neq \text{chr}(r, A)$, then $(2^A - 3^r) \nmid B(v)$.*

The hypothesis $v \neq \text{chr}$ is needed only for $A = 2r$. There an identity “swap” of equal entries returns the word $(2, \dots, 2)$ of the trivial cycle $x = 1$.

Theorem 3 (Classification, NormAll.knight_all, one_move_classification). *For $r \geq 2$ and $3^r + 1 < 2^A$: $q \mid B(\text{chr}(r, A))$ if and only if $A = 2r$. More generally, for v equal to $\text{chr}(r, A)$ or one move from it, $q \mid B(v)$ holds exactly when $A = 2r$ and $v = \text{chr}(r, A)$.*

Corollary 4 (Bridge, NormBridge.cycle_one_move_christoffel). *Suppose a positive T -cycle, read from an odd point m , has length L and valuation word v with $r \geq 2$ odd steps. If v equals $\text{chr}(r, L)$ or is one move from it, then $m = 1$.*

2.1 Proof in the coprime case

Assume $\gcd(A, r) = 1$, and write $\rho_j = jA \bmod r$.

Step 1: a distinguished unit (NormReduce.exists_g). Choose u, k with $ru - Ak = 1$ and set $g = 2^u 3^{-k} \in \mathbb{Z}/q$. Then $g^r = 2$ and $g^A = 3$, and g is unique with these properties. Each term of $B(\text{chr})$ is a power of g : $3^{r-1-j} 2^{\lfloor jA/r \rfloor} = G g^{r-1-\rho_j}$ with $G = g^{(A-1)(r-1)}$. Since $j \mapsto \rho_j$ is a permutation, summing gives the identity

$$(g - 1) B(\text{chr}(r, A)) = g^{(A-1)(r-1)} \quad \text{in } \mathbb{Z}/q \quad (\text{knight_identity}).$$

The right-hand side is a unit, so $B(\text{chr})$ is a unit modulo q , i.e. $\gcd(B(\text{chr}), q) = 1$. This recovers Knight's theorem. The device is not new: Mghirbi's ξ is g^{-1} and he gives the same three-line argument. Lebel's radius-0 theorem uses the same permutation and geometric sum, prime by prime. The only difference here is that we work modulo the full, possibly composite, q .

Step 2: two trinomials (NormReduce.reduce). A move of chr away from the wrap-around position changes exactly one partial sum by ± 1 , and hence changes B by a single term. This local identity is Lebel's Flip Lemma 5.1; the prefix case also appears in [7]. A wrap-around move shifts all of $A_1, \dots, A_{r-1}$ at once. For these, the identities $B(v) + 3^{r-1} = 2B(w)$ or $2B(v) = B(w) + 3^{r-1}$ again express $B(v)$ through $B(w)$ by a single term. In every case, combined with Step 1, $(g - 1)B(v)$ is a unit times a trinomial in g . So $q \mid B(v)$ forces g to be a root of

$$\text{Type I: } g^{p+1} - g^p + 1 = 0 \quad (1 \leq p \leq r-1), \quad \text{Type II: } g^e - g + 1 = 0 \quad (2 \leq e \leq r).$$

(The values $p = 0$ and $e = 1$ give a nonzero unit and are excluded.) When $A < 2r$ the exponents satisfy $p + 1, e \leq A - r$. Type II corresponds to left flips and Type I to right flips. The reduction is Lebel's Theorem 6.3, carried out modulo q rather than modulo a prime factor.

Step 3: from a modular root to an integer (NormDet). Write the trinomial as $g^\alpha - g^\beta + 1$, so $(\alpha, \beta) = (p + 1, p)$ for Type I and $(e, 1)$ for Type II. Let S_n be the $r \times r$ "shift with weight 2 on wrap-around" matrix, so that $(1, g, \dots, g^{r-1}) S_n = g^n (1, g, \dots, g^{r-1})$ because $g^r = 2$. For $1 \leq \alpha \neq \beta < r$, a trinomial root gives a row vector annihilated by $M = S_\alpha - S_\beta + I$ modulo q whose first entry is 1. The adjugate then gives $q \mid \det M$ (dvd_det_of_vecMul). Since M is unitriangular modulo 2, $\det M$ is odd (det_three_odd). Hence $\det M \neq 0$, so $q \leq |\det M|$. AM-GM on the eigenvalues of $M^T M$ gives $\det(M)^2 r^r \leq \|M\|_F^{2r}$ (det_sq_mul_1e), and $\|M\|_F^2 = 3r + 3\alpha + 3\beta$ (frob_three). Hence

$$q^2 r^r \leq (3r + 3\alpha + 3\beta)^r.$$

The restriction $\alpha, \beta < r$ is why the two edge exponents are treated separately below.

Step 4: assembly (`NormMain.main`).

- *Case* $2^A > 2 \cdot 3^r$. Then $q > 3^r$. On the other hand $\alpha, \beta < r$ gives $(3r + 3\alpha + 3\beta)/r < 9$, so Step 3 gives $q < 3^r$, a contradiction. The two edge exponents, $p = r - 1$ and $e = r$, are handled directly: they give $q \mid 2^{A+1-r} - 1$, respectively $q \mid 3^r - 2$, which is impossible.
- *Case* $2^A \leq 2 \cdot 3^r$. This case does not occur for $r = 2$. Here $A = \lceil r \log_2 3 \rceil$ and $A < 2r$, so the exponents are at most $A - r \approx 0.585r$, and Step 3 gives $q \lesssim \sqrt{3 + 6 \cdot 0.585^r} \approx 2.55^r$.
 - For $r \geq 150$ this contradicts the gap lemmas of the library: $2^A \leq 2^{30}q$ for $r < 2.26 \cdot 10^8$ (`gapBelow_225644606`, from the continued fraction of $\log_2 3$), and $2^A \leq 2^{172}r^{58}q$ beyond (`gap_all159`, from a formally verified effective irrationality measure).
 - For $3 \leq r < 150$, a kernel-checked table finds no trinomial root at the unique g (`NormFinite.finite_check`).

2.2 The case $\gcd(A, r) = d \geq 2$

Following Solomon (Prop. 6.3), write $r = dr'$, $A = dA'$, $U = 2^{A'}$, $V = 3^{r'}$ and $S = \sum_{s < d} U^s V^{d-1-s}$. Then:

- $q = (U - V)S$, with $S > 1$ and $\gcd(S, 6) = 1$;
- $\text{chr}(r, A)$ is the d -fold repetition of $\text{chr}(r', A')$, so $B(\text{chr}) = S \cdot B(\text{chr}(r', A'))$ (`solomon_factor`);
- since the entries of chr differ by at most one, a move changes B by a single monomial $\pm 2^x 3^y$, and S cannot divide a monomial (`one_move_not_dvd`).

Wrap-around moves are handled by the identities $B(v) + 3^{r-1} = 2B(w)$ or $2B(v) = B(w) + 3^{r-1}$, so no rotation argument is needed.

3 Extensions

All statements below are unconditional and Lean-verified. They are stated at word level, $q \nmid B(v)$, and, where indicated, in cycle form.

Proposition 5 (Parameters of actual cycles, `NormCycleAll`). *Every nontrivial positive T -cycle has $L < 2r$ and $r \geq 40901$ (`cycle_params`), and in fact $200L < 317r$ (`NormGeom`). It also has an odd point (`exists_odd_point`).*

The constant 40901 is assembled from existing library lemmas. A nontrivial C -cycle has length $L + r \geq 122703$ (a formalized Crandall–Eliahou-type bound), and $L < 2r$ because $3^r < 2^L \leq 3^r(1 + \frac{1}{3m})^r < 4^r$, where m is the cycle minimum. Hence $3r > 122703$, so $r \geq 40902$; the Lean statement records the slightly weaker $r \geq 40901$. The constant is far from sharp. The published literature gives much more: Hercher’s bounds [8], combined with Barina’s verification to 2^{71} [9], give $r > 1.375 \times 10^{11}$. Our constant is only the one that was convenient to formalize, and it suffices for every theorem below.

Every actual cycle therefore satisfies $r \geq 40901$ and $A < 2r$. So the hypotheses $r \geq 2325$, $r \geq 40901$ and $A < 2r$ in the results below are automatically met when the results are applied to cycles. Coprimality, where it is assumed, is not automatic.

Theorem 6 (A unit slid any distance, NormShiftSlide). *Let $r \geq 40901$ and $3^r + 1 < 2^A$, with any $\gcd(A, r)$. Move one unit from position i_0 (where $\text{chr}_{i_0} \geq 2$) to an arbitrary position $i_1 \neq i_0$. The resulting word v satisfies $q \nmid B(v)$ (no_cycle_slide_allRA). Consequently, no nontrivial positive T -cycle, read from its chosen odd start, has such a valuation word (cycle_slide_all).*

Method (NormShift): a shift lift. Let P_i be the partial sums of the periodic extension of v , and $G_a = \sum_{j < r} 3^{r-1-j} 2^{P_{a+j}}$ the numerator of the a -th rotation. If $q \mid B(v)$, then q divides every G_a , and each G_a is 2^{P_a} times an odd number. Choose a shift $\sigma < r$ and write $\sigma A = rm + t$ with $1 \leq t < r$. For a suitable integer $H \geq 0$, the combination $2^H G_a - 2^{H+m} G_{a-\sigma}$ is a sum of terms indexed by the positions where the deviation of v from chr is not shift-invariant. If these positions leave a long cyclic window free, a size estimate against the gap lemma, together with the oddness of the G_a , gives a contradiction. For $t = 1$ in the coprime case this is Mghirbi's lift (Thm. 6.3). What is new is that it needs no distinguished root, works for any t and any $\gcd$, and uses windows shaped as arcs.

The same shift-lift argument gives a run-count criterion (NormShift.few_levels_coprime; cycle form NormShiftSlide.cycle_few_levels_coprime). Write $\varepsilon_i = A_i - \lfloor iA/r \rfloor$ for the deviation of the partial sums from the Christoffel line. Suppose a coprime word has $\max_i |\varepsilon_i| \leq H_0$, and its deviation changes value at only J cyclic positions. Then $q \nmid B(v)$ whenever $2^{2H_0+174} r^{59} < 3^{\lfloor r/(6J+3) \rfloor + 1}$. This complements Mghirbi's support-height criterion.

Theorem 7 (Two and a few flips, coprime). *Let $\gcd(A, r) = 1$ and $3^r + 1 < 2^A$. In items 1–2 assume also $r \geq 2325$ and $A < 2r$. A flip at site $k \in (0, r)$ means that the partial sum A_k of chr is lowered (left flip) or raised (right flip) by one. The Lean statements carry the corresponding site conditions, which say that the flip is legal, i.e. all entries stay ≥ 1 . The following words v satisfy $q \nmid B(v)$, and the corresponding cycle forms hold.*

1. *Two left flips (NormTwo.no_cycle_two_left_flips). For cycles this is Mghirbi's $E = 2$; the formalization is ours.*
2. *Any two right flips, including adjacent ones and the wrap-around case $A = 2r - 1$ (NormFlips.no_cycle_two_right_flips_all). This is extended to every $\gcd$ in NormCofactor.no_cycle_two_right_flips_allRA.*
3. *For $r \geq 40901$: right flips at a set K of sites, with exponents $p_k = r - 1 - (kA \bmod r)$ satisfying $5 \leq p_k$, $p_k + 6 \leq r$, $|p_k - p_{k'}| \geq 6$ for $k \neq k'$, and $\sum_{k \in K} 4^{p_k/r} \leq 29/5$ (NormPoly.no_cycle_k_right_flips_sep).*
4. *Mixed pairs, one left and one right flip, in an explicit positive-density region of pair space (NormMixed).*

These results rest on weighted Hadamard bounds for the companion-type matrix of the perturbation polynomial. The bounds are sharpened by a unitriangular column operation, which we call a “Szegő filter” by analogy with one-step prediction filters (NormSparse, NormFilter, NormPoly).

Theorem 8 (Non-coprime families, $d = \gcd(A, r) \geq 2$).

1. Every perturbation of the partial sums of chr by ± 1 at two sites, with any signs and at any distance (`NormCofactor.no_cycle_two_site_noncoprime`).
2. Deviations confined to one window of length about $(1 - 1/d)r$ (`NormWindow`), or to a cyclic arc (`NormArc`).
3. Deviations at up to $d - 1$ arbitrary sites (`NormArc`).
4. A unit slid any distance, for $d \geq 3$ (`NormPlateau`). Theorem 6 covers $d = 2$ and $d = 1$ as well.
5. Self-fold (`NormFold`). Let $p \geq 2$ be any common factor of A and r , and write $r = pr'$. Let v be a word whose partial sums stay within e of those of chr , differ from them only on a set D of sites, and are not r' -periodic. Then $q \nmid B(v)$, provided $|D|\gamma < r'$ for an integer γ with $(r'2^{4e+3})^2 < 2^{3(\gamma+1)}$ (this is the condition for $p = 2$; the general- p condition has the same shape), so $\gamma \approx \frac{2}{3} \log_2 r' + \frac{8}{3}e$. The proof folds $B(v)$ against its own r' -shift modulo $\Phi_p(2^{A/p}, 3^{r/p})$. For example, at $r = 40902$, $A = 64830$ ($d = 6$) it allows about 1460 sites, compared with $d - 1 = 5$ in item 3.

4 Limits and status

- *Not a cycle exclusion in general.* Every result concerns words near the Christoffel word, while typical words are far from it. The norm and determinant bounds stop at $O(1)$ moves. Stacked moves, where one partial sum changes by 2, and the remaining mixed pairs are open. The shift-lift method reaches about $r/\log r$ level changes, which is still a thin class.
- *Rotations.* The slide theorem is stated for slides of chr read from position 0. Since divisibility is rotation-invariant, re-reading a cycle from another odd point covers slides of rotations, but that step is not formalized. The one-move theorem includes cyclic moves, so it needs no such step.
- *Lebel's single-prime form.* Write $D = q$ and $P^+(D)$ for its largest prime factor. The single-prime form is proved only conditionally, when $P^+(D) \gtrsim 16r D^{2/3}$ (`NormLebel`), and in one composite instance, $(A, r) = (37, 19)$ with $D = 5 \cdot 27255338401$ (`NormPlateauLift`).
- *Novelty.* A preliminary literature search found no source for Theorem 2. Its left-flip half is known for actual cycles (Mghirbi, Solomon). The Christoffel case is Knight's for $\gcd = 1$ and Solomon's (Prop. 6.2) for $\gcd > 1$, again for actual cycles. The distinguished unit and the three-line Knight argument are Mghirbi's. The trinomial reduction and the local flip identity are Lebel's. Lebel's closure of the canal $s = 2$ covers some right flips. What appears new is the rest of the right-flip half, the uniform word-level divisibility statement for all (r, A) , and the determinant proof that works modulo the full q . The primary texts of Lebel, Mghirbi and Solomon should still be checked line by line. At cycle level, several of the lift results reduce to Terras's 2-adic separation [10] plus a bound on the maximum of the cycle, and should be regarded as known in substance.

Lean index

Result	Lean name	File
Knight identity	<code>knight_identity</code> , <code>knight</code>	<code>NormReduce</code>
Trinomial reduction	<code>reduce</code>	<code>NormReduce</code>
Determinant bound	<code>le_typeI</code> , <code>le_typeII</code>	<code>NormDet</code>
Small r table	<code>finite_check</code>	<code>NormFinite</code>
Theorem 2, coprime	<code>no_cycle_one_move_christoffel</code>	<code>NormMain</code>
Theorem 2, all (r, A)	<code>no_cycle_one_move_christoffel_all</code>	<code>NormAll</code>
Classification	<code>knight_all</code> , <code>one_move_classification</code>	<code>NormAll</code>
Bridge to cycles	<code>cycle_one_move_christoffel</code>	<code>NormBridge</code>
Run-count criterion	<code>few_levels_coprime</code>	<code>NormShift</code>
Two/few flips	<code>no_cycle_two_left_flips</code> , <code>no_cycle_k_right_flips_sep</code> , ...	<code>NormTwo</code> , <code>NormFlips</code> , <code>NormPoly</code>
Non-coprime families	...	<code>NormCofactor</code> , <code>NormArc</code> , <code>NormPlateau</code> , <code>NormFold</code>
Cycle parameters	<code>cycle_params</code>	<code>NormCycleAll</code>
Slide any distance	<code>no_cycle_slide_allRA</code> , <code>cycle_slide_all</code>	<code>NormShiftSlide</code>

All files are in `claude-search/lean/CollatzSearch/`.

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